Chemistry - Graduate Level Assessment 21 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. The executioner in charge of the lethal gas chamber at the state penitentiary adds excess dilute H_2 SO_4 to 196 g (about (1/2) lb) ofNaCN. What volume of HCN gas is formed at STP?

   (a) 11.2 liters
   (b) 67.2 liters
   (c) 168 liters
   (d) 100.8 liters
   (e) 22.4 liters

2. Calculate the average separation of the electron and the nucleus in the ground state of the hydrogen atom.

   (a) 0.159 pm
   (b) 79.5 pm
   (c) 210 pm
   (d) 132 pm
   (e) 0.53 pm

3. Calculate the number of $H^+ ionsintheinnercompartmentofasingleratlivermitochondrion, assuming$

   4. 1000 $H^+ ions$

   4. 900 $H^+ ions$

   4. 843 $H^+ ions$

   4. 950 $H^+ ions$

   4. 615 $H^+ ions$

Which is correct about the calcium atom?

(a) It contains 20 protons and neutrons
(b) All atoms of calcium have 20 protons and 40 neutrons
(c) It contains 20 protons, neutrons, and electrons

(d) All calcium atoms have 20 protons, neutrons, and electrons

(e) All atoms of calcium have a mass of 40.078 u

In which of the following is not the negative end of the bond written last?

(a) H-O

(b) P-Cl

(c) N-H

(d) S-O

(e) C-O

True / False (10 marks)

Answer True or False

6. True or False: Silver (Ag) possesses a higher electronegativity than Iron (Fe) due to its position in the periodic table.

7. True or False: The volume of the rectangular tank is 250 liters when calculated from its dimensions of 2.0 m, 50 cm, and 200 mm.

8. True or False: CBr 4 has a tetrahedral molecular geometry due to its four equivalent bonding pairs of electrons around the central carbon atom.

9. True or False: The conjugate acid of the H_2PO_4- ion is H_4PO_4-.

10. True or False: The frequency of light emitted by a helium-neon laser at 632.8 nm is 5.750 x $10^1$4 Hz.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate the mass of calcium hydroxide required to neutralize 28.0 g of hydrochloric acid, and then determine the mass of phosphoric acid that this amount of calcium hydroxide can neutralize. Discuss the underlying chemical reactions and stoichiometric principles involved in your answer.

12. Discuss the relationship between the chemical potentials of components in a binary mixture of water and ethanol, given a mole fraction of water at 0.60 and a change in water's chemical potential. How does this affect the chemical potential of ethanol?

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